

Online Food Delivery Analysis

1. Project Overview

Online food delivery platforms generate large volumes of transactional and operational data. Analysing this data helps businesses understand customer behaviour, delivery performance, restaurant efficiency, and revenue trends. This project aims to perform end-to-end data analysis using Python, SQL, and Power BI to extract actionable business insights from food delivery operations.

2. Problem Statement

The objective of this project is to analyse online food delivery data and identify key factors influencing revenue, customer engagement, delivery efficiency, and restaurant performance. The project also aims to create an interactive dashboard for business decision-making.

3. Objectives

- Perform data profiling and quality assessment.
- Clean and preprocess the dataset.
- Engineer meaningful business features.
- Store and analyse data using SQL.

- Conduct Exploratory Data Analysis (EDA).
- Build interactive Power BI dashboards.
- Generate actionable business recommendations.

4. Dataset Overview

Dataset Size: 100,000 Records

Key Attributes:

- Customer Information
- Restaurant Information
- Order Details
- Delivery Metrics
- Payment Information
- Ratings and Reviews
- Profit Margin Data

Total Columns: 25 Original Columns

Additional Features Created:

- Age Group
- Delivery Category
- Revenue Category

- Profit Category
- Delivery Rating Filled

5. Data Profiling

Initial profiling identified:

- Missing values across multiple columns.
- High missing percentage in Final Amount and Customer Age.
- Missing delivery ratings and cancellation reasons.
- Presence of negative final amounts caused by excessive discounts.
- No major duplicate records detected.

Data quality checks included:

- Missing value analysis
 - Duplicate detection
 - Outlier analysis
 - Data type verification
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6. Data Cleaning and Feature Engineering

Data Cleaning

- Customer Age imputed using median values.
- Missing categorical values replaced using mode.
- Final Amount recalculated using:

Final Amount = Order Value - Discount Applied

- Negative final amounts corrected to zero.
- Date fields converted into proper datetime format.

Feature Engineering

Created business-focused features:

1. Age Group
2. Delivery Category
3. Revenue Category
4. Profit Category
5. Delivery Rating Filled

These features improved analytical capability and dashboard usability.

7. SQL Database Implementation

The cleaned dataset was loaded into MySQL for relational database analysis.

SQL Operations Performed

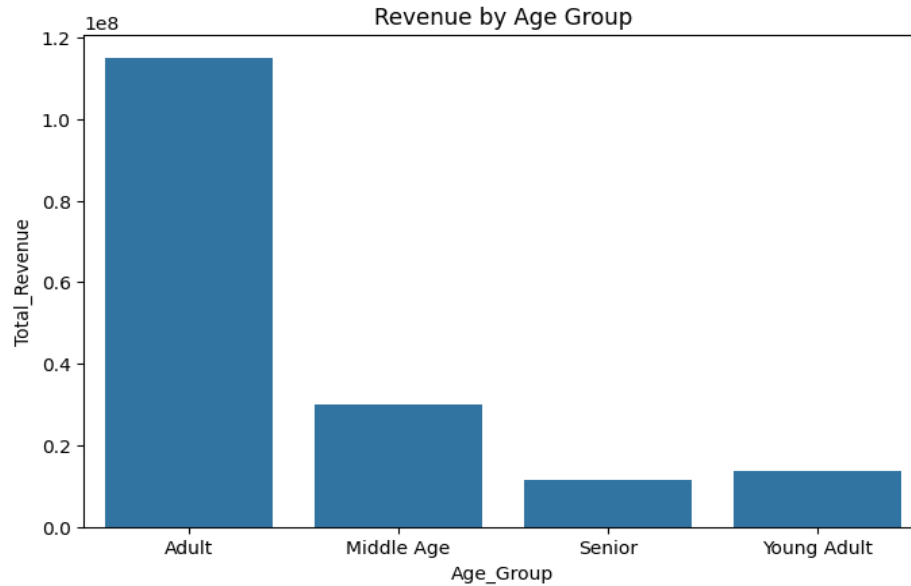
- Database Creation
- Table Creation
- Data Import
- Aggregation Queries
- Business KPI Analysis
- Customer Analysis
- Revenue Analysis
- Restaurant Analysis

A total of 15 SQL business queries were developed to support operational and strategic decision-making.

8. Exploratory Data Analysis

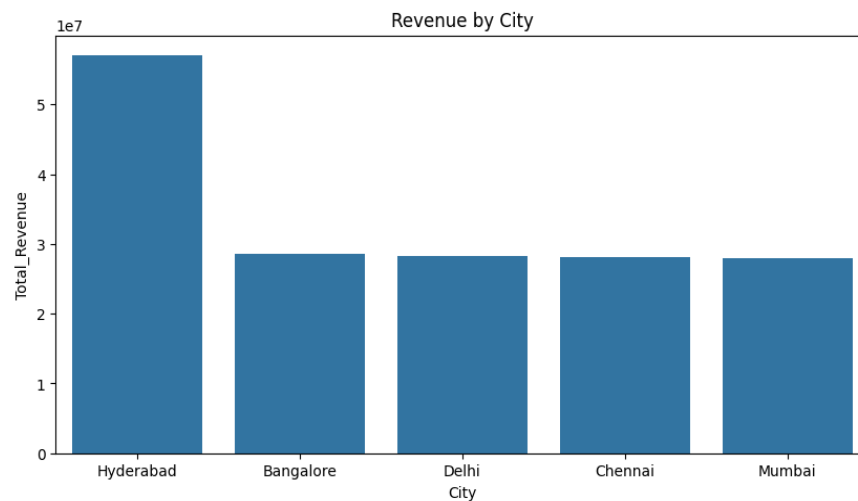
Customer Analysis

- Adult customers generated the highest revenue.
- Customer spending was relatively balanced across age groups.
- Top customers contributed a small percentage of total revenue, indicating a diversified customer base.



Revenue Analysis

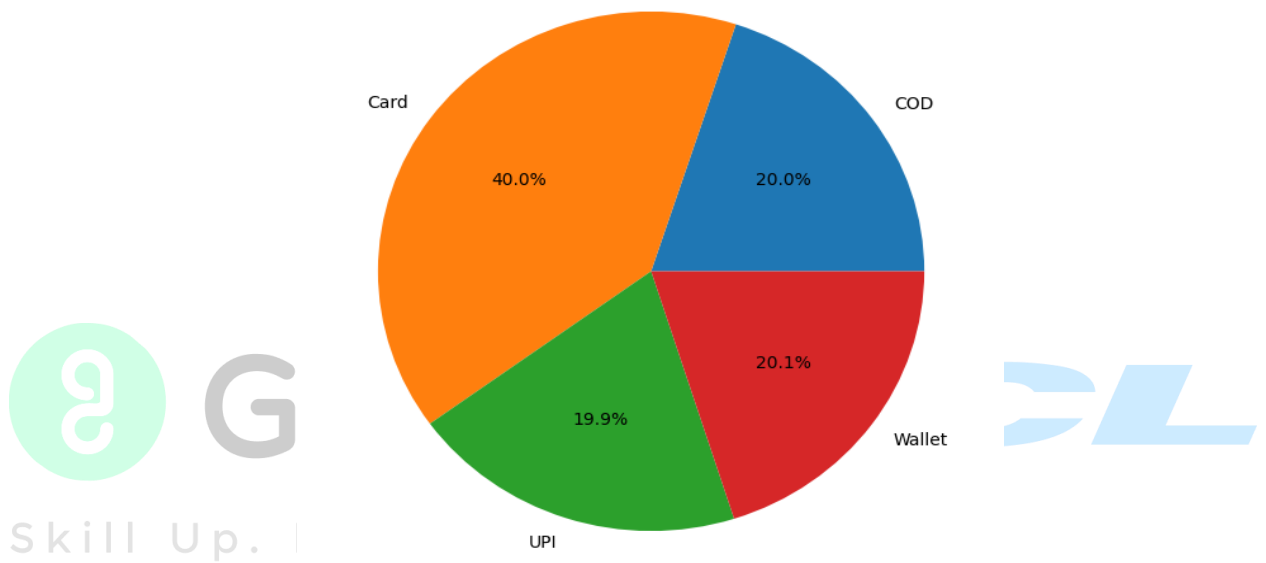
- Total Revenue: Approximately 170 Million
- Revenue remained stable throughout the year.
- Hyderabad generated the highest revenue among all cities.



Payment Mode Analysis

- Card payments generated the highest revenue.
- UPI, Wallet, and COD showed similar revenue contributions.

Order Share by Payment Mode

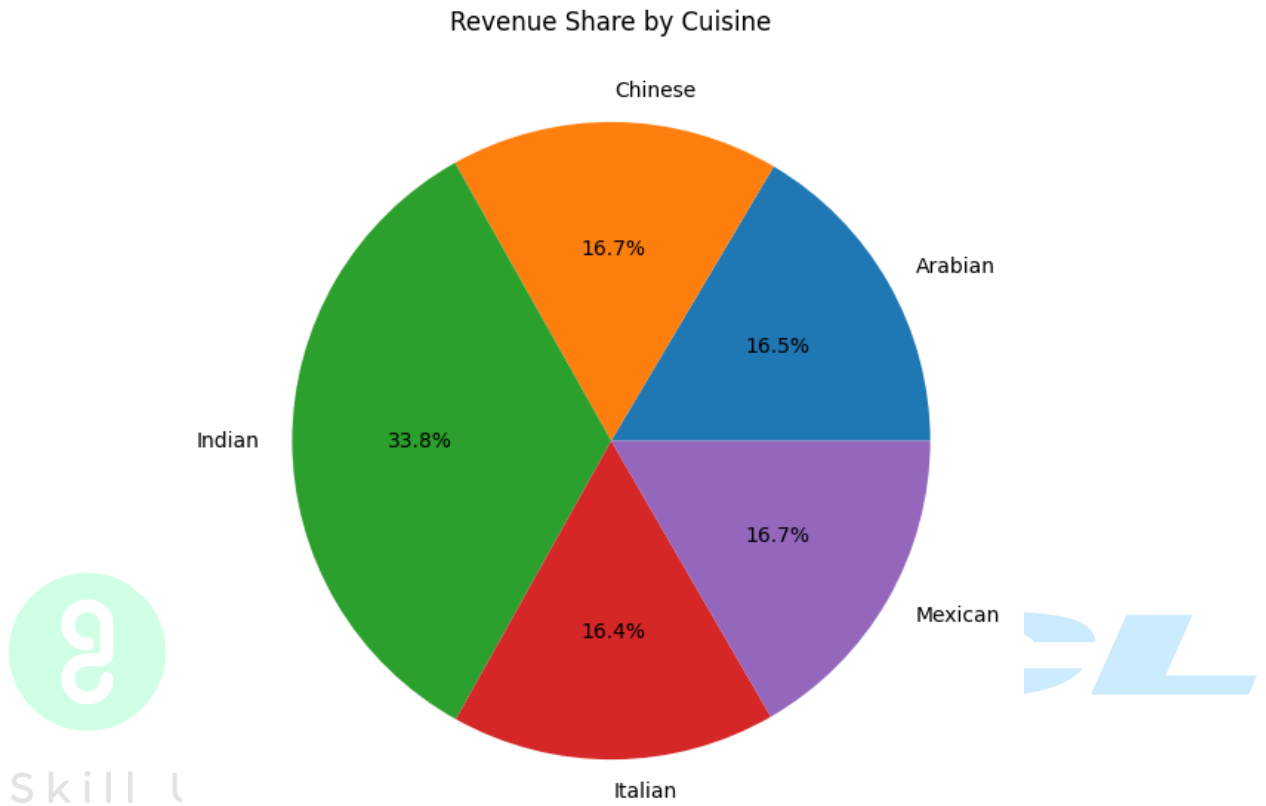


Cuisine Analysis

Revenue Ranking:

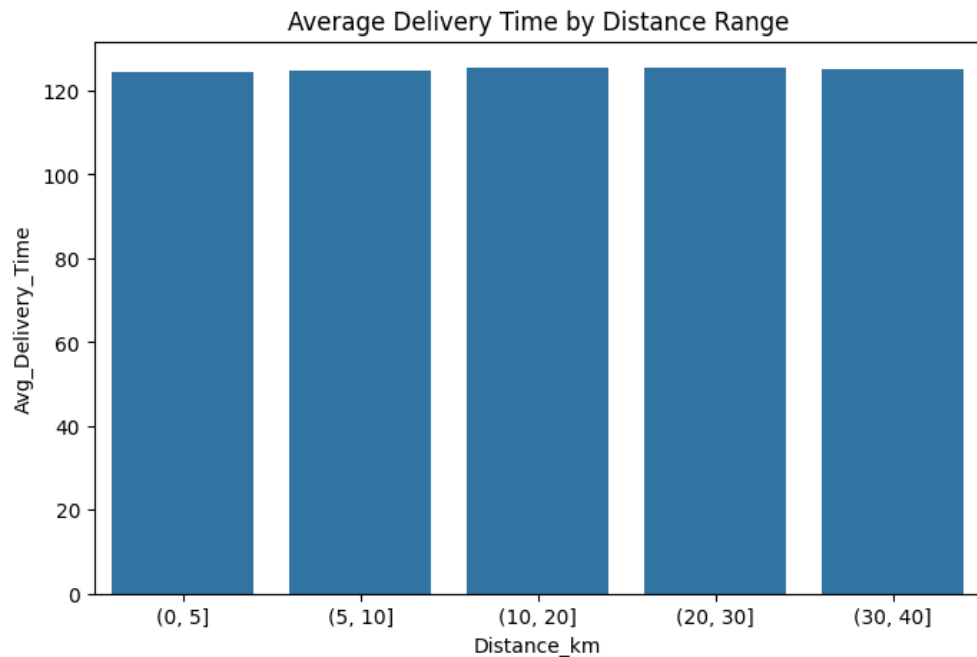
1. Indian Cuisine
2. Mexican Cuisine
3. Chinese Cuisine
4. Arabian Cuisine
5. Italian Cuisine

Indian cuisine emerged as the strongest revenue contributor.



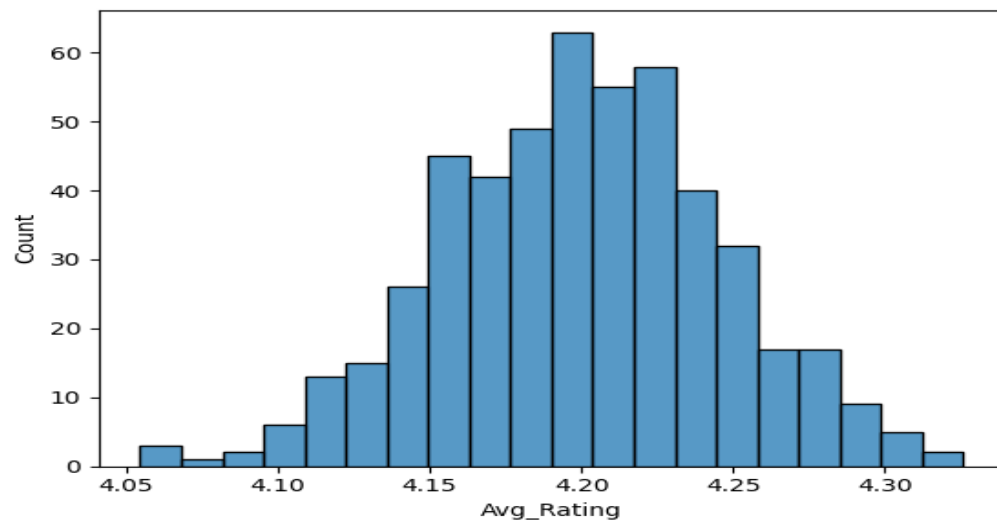
Delivery Analysis

- Average Delivery Time: ~125 minutes
- Delivery performance remained consistent across cities.
- Peak-hour operations handled approximately one-third of total orders.



Restaurant Analysis

- Several restaurants consistently maintained ratings above 4.3.
- Cancellation rates varied across restaurants.
- Restaurant quality directly influenced customer satisfaction

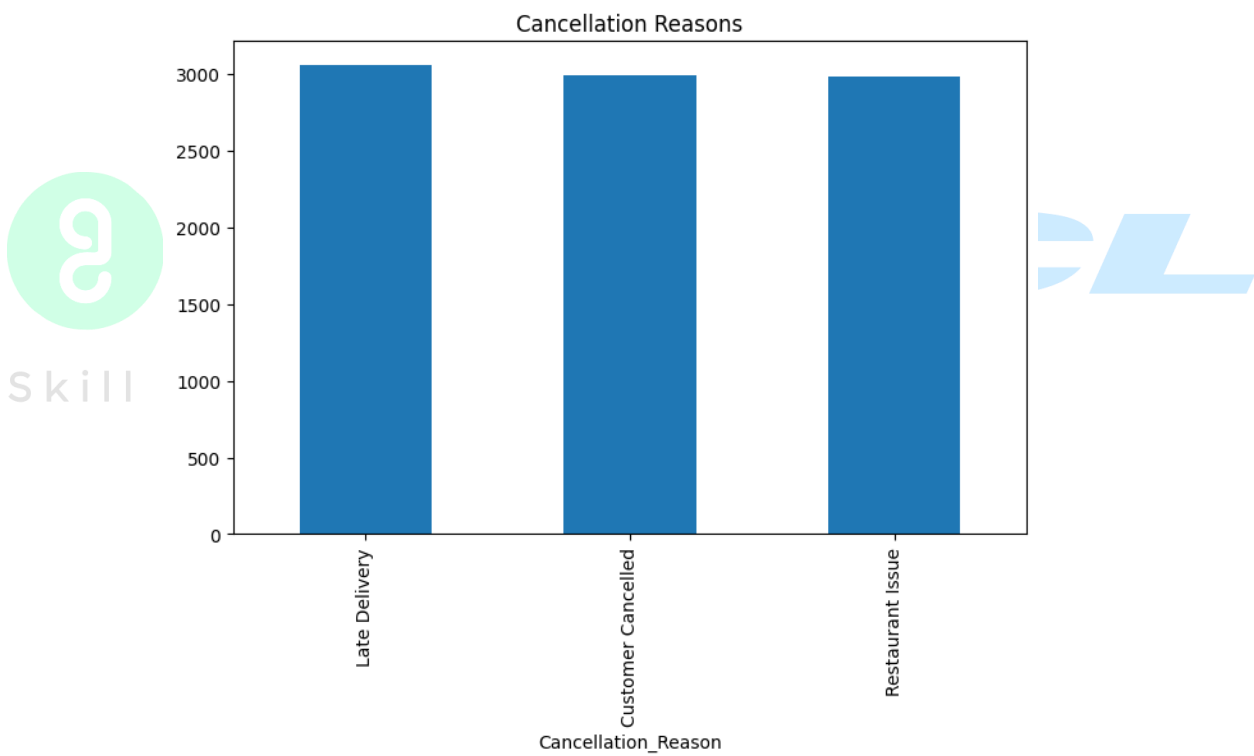


Cancellation Analysis

Primary cancellation reasons:

- 1. Late Delivery
- 2. Customer Cancelled
- 3. Restaurant Issue

Late delivery emerged as the leading operational challenge.

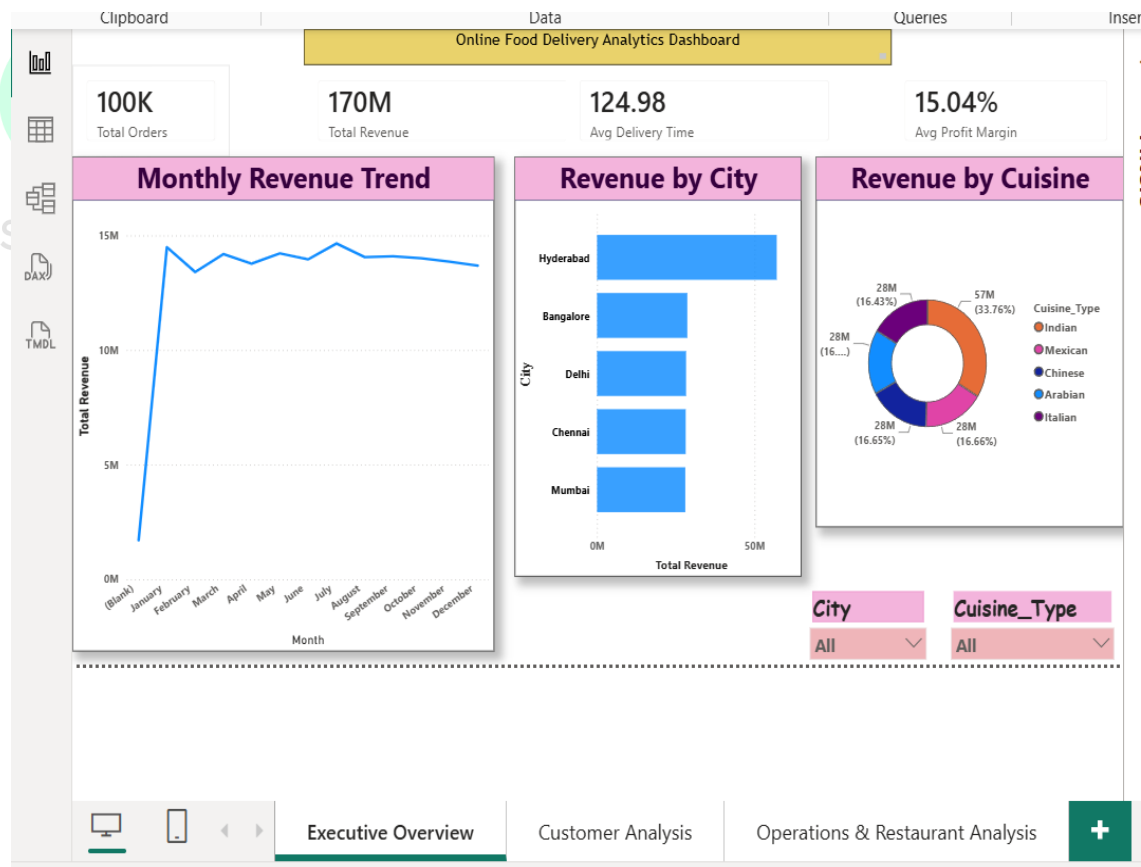


9. Power BI Dashboard

Three interactive dashboard pages were developed:

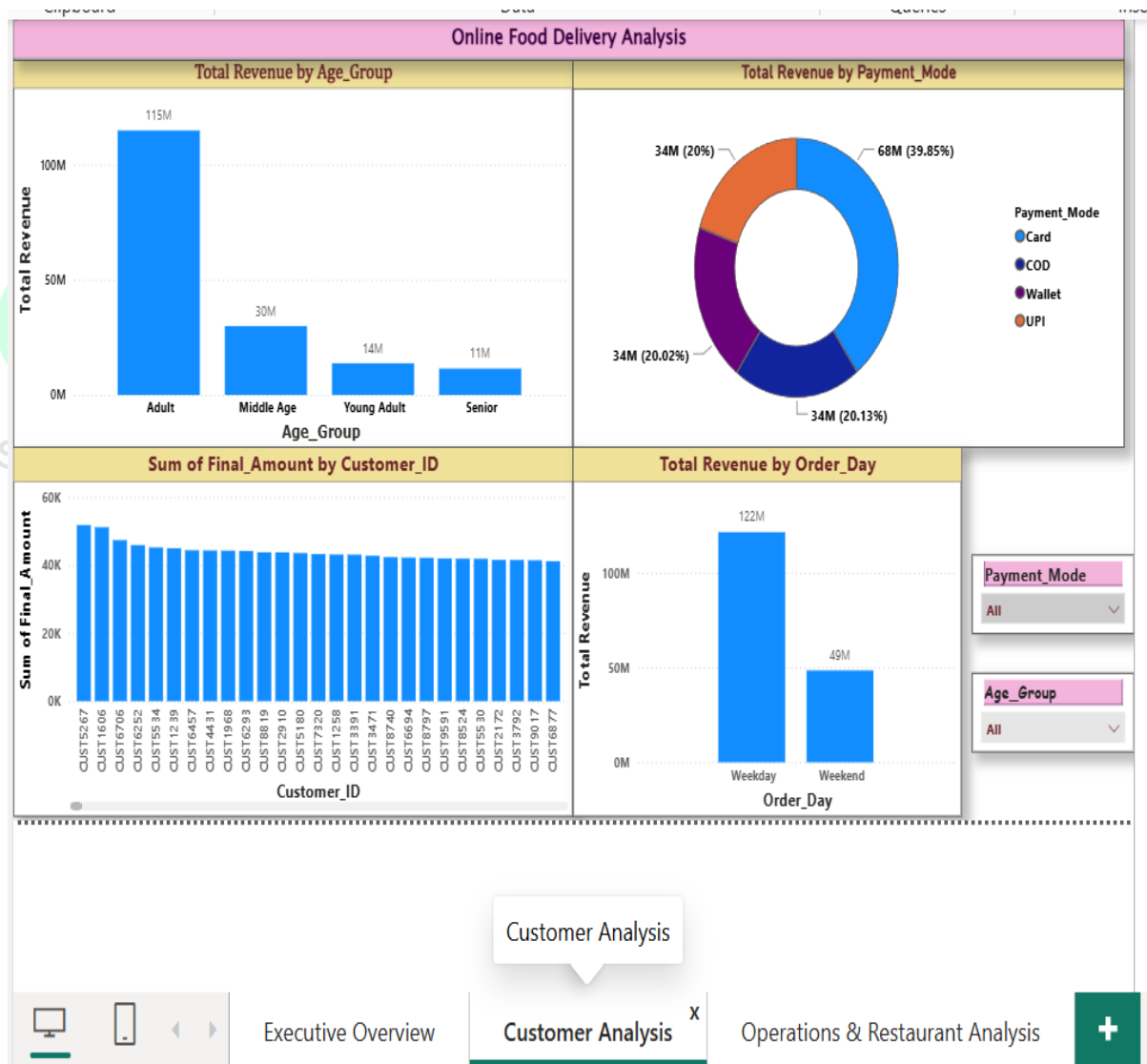
Executive Overview

- KPI Cards
- Monthly Revenue Trend
- Revenue by City
- Revenue by Cuisine
- Interactive Filter



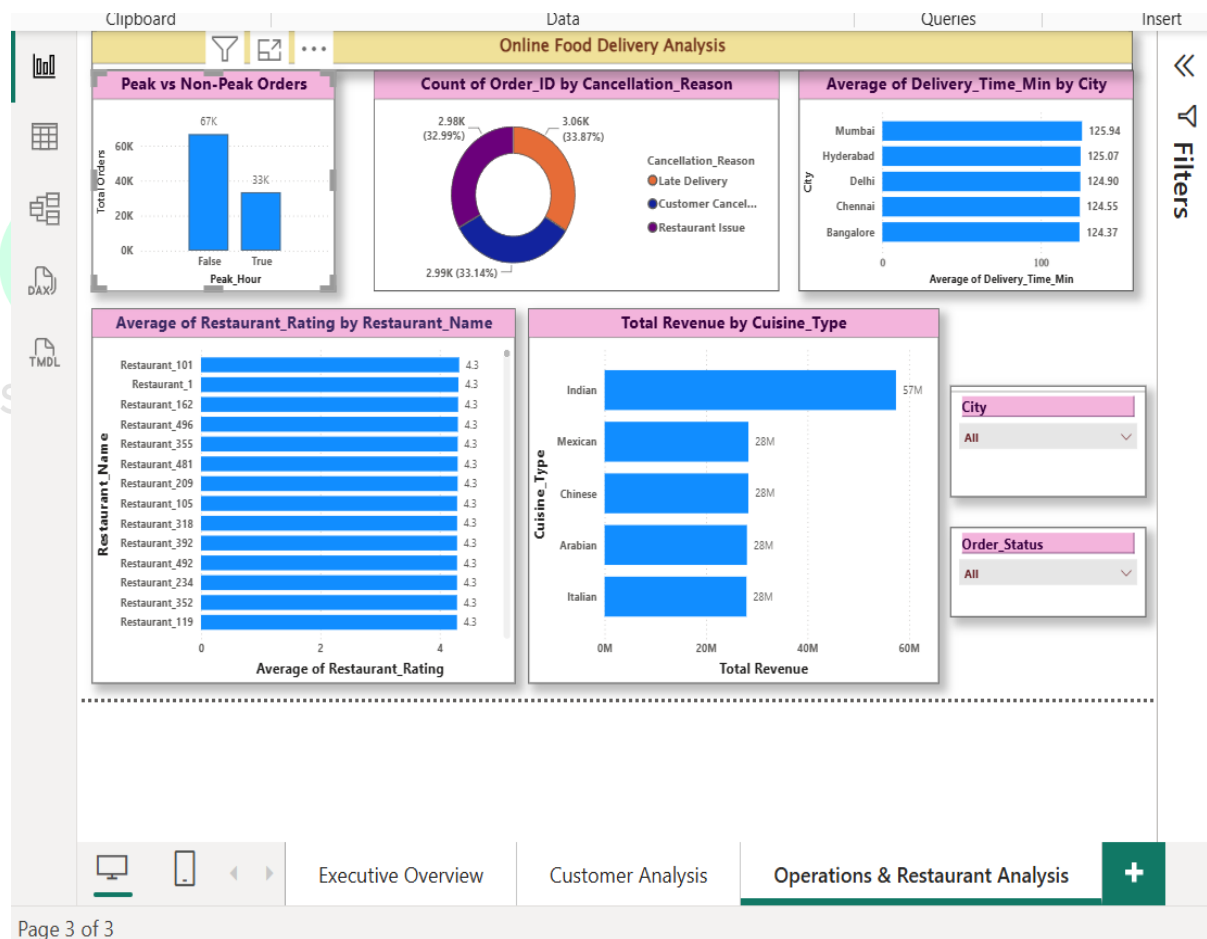
Customer Analysis

- Revenue by Age Group
- Payment Mode Analysis
- Top Customers
- Weekday vs Weekend Analysis



Operations & Restaurant Analysis

- Peak Hour Analysis
- Cancellation Analysis
- Delivery Performance
- Top Rated Restaurants
- Cuisine Revenue Analysis



10. Key Insights

1. Hyderabad generated the highest revenue among all cities.
2. Indian cuisine accounted for the largest revenue share.
3. Adult customers formed the largest customer segment.
4. Card payments contributed the highest revenue.
5. Revenue remained stable throughout the year.
6. Peak-hour orders represented approximately one-third of total orders.
7. Delivery performance was consistent across cities.
8. Late delivery was the most common cancellation reason.
9. Restaurant ratings remained generally high across the platform.
10. Customer spending was evenly distributed without heavy dependence on a few customers.

11. Business Recommendations

Recommendation 1: Improve Delivery Efficiency

Late delivery was identified as the leading cancellation reason. Optimizing route planning, delivery allocation, and real-time tracking can significantly reduce cancellations.

Recommendation 2: Strengthen Operations During Peak Hours

Since a significant portion of orders occur during peak periods, additional delivery partners and operational support should be allocated during these hours.

Recommendation 3: Expand High-Performing Cuisine Categories

Indian cuisine consistently generated the highest revenue. Marketing efforts and restaurant partnerships should focus on expanding this segment.

Recommendation 4: Leverage Card Payment Promotions

Card payments contributed the highest revenue. Strategic cashback and loyalty programs can further improve customer retention.

Recommendation 5: Focus on High-Revenue Cities

Hyderabad demonstrated the strongest revenue performance. Similar operational strategies can be replicated in other cities to improve overall revenue.

Recommendation 6: Monitor Restaurant Cancellation Rates

Restaurants with higher cancellation rates should be reviewed and provided operational guidance to improve service quality.

12. Conclusion

This project successfully implemented a complete data analytics workflow using Python, SQL, and Power BI. The analysis revealed important customer, operational, and revenue insights that can support business decision-making. The developed dashboards provide an interactive platform for monitoring performance and identifying growth opportunities within online food delivery operations.